

RAG Retrieval & Evaluation Metrics

Interview-focused guide: retrieval quality, generation quality, RAGAS, numerical examples, Python snippets, and production drift monitoring.

1. The RAG Evaluation Mental Model

Retrieval evaluation asks: “Did we retrieve the right evidence?”

Generation evaluation asks: “Did the LLM use that evidence correctly and answer the user?”

Production monitoring asks: “Is the system still performing well after deployment and data/model changes?”

2. Retrieval Metrics

Metric	What it measures	Interview interpretation
Precision@K	How many of the top K retrieved chunks are relevant.	Are we returning too much irrelevant context?
Recall@K	How much of the relevant evidence was retrieved in top K	Did we miss important information?
MRR	How high the first relevant result appears.	Is the first useful chunk ranked near the top?
NDCG@K	Ranking quality with graded relevance; highly relevant items get more weight	Are the most relevant results ordered correctly?

Precision@K

Formula: $\text{Precision@K} = \text{relevant retrieved chunks in top K} / K$.

Example: Top 5 results contain 4 relevant chunks → $\text{Precision@5} = 4/5 = \mathbf{0.80 (80\%)}$.

Recall@K

Formula: $\text{Recall@K} = \text{relevant retrieved chunks in top K} / \text{total relevant chunks available}$.

Example: There are 6 relevant chunks in the corpus and the system retrieves 4 of them in top 5 → $\text{Recall@5} = 4/6 = \mathbf{0.67 (67\%)}$.

MRR — Mean Reciprocal Rank

For one query, reciprocal rank = $1 / \text{rank of the first relevant result}$. Across queries, average those values.

Example: first relevant result is #2 → reciprocal rank = $1/2 = 0.50$. If three queries have first relevant ranks 1, 2, and 4, $\text{MRR} = (1 + 0.5 + 0.25)/3 = \mathbf{0.583}$.

NDCG@K

NDCG is useful when relevance is graded rather than simply relevant/not relevant. For example, a result can be highly relevant, somewhat relevant, or irrelevant. It rewards putting highly relevant documents earlier.

3. Retrieval vs Generation Metrics

Layer	Typical metrics	Question answered
Retriever	Precision@K, Recall@K, MRR, NDCG@K	Did we retrieve and rank useful evidence?
Generator	Faithfulness, Answer Relevancy	Did the answer stay grounded and address the question?
Context	Context Precision, Context Recall	Was the supplied context relevant and sufficiently complete?
Production	Latency, cost/request, feedback, failure rate	Is the system healthy and useful in real usage?

4. Faithfulness / Groundedness

Faithfulness measures whether the generated answer is supported by the retrieved context. A fluent answer can still be unfaithful if it invents information not present in the evidence.

Interview example: Context says “The refund period is 30 days.” If the model answers “The refund period is 60 days,” the answer is not faithful even if it sounds confident.

5. Answer Relevancy

Answer relevancy checks whether the generated response actually addresses the user's question. An answer can be factually grounded but still irrelevant—for example, giving a long explanation of authentication when the user asked for the refund deadline.

6. Context Precision and Context Recall

Context Precision: Are the retrieved chunks mostly useful? High precision means little irrelevant context is being passed to the LLM.

Context Recall: Did retrieval capture the information needed to answer the question? Low recall often means the correct evidence was missed.

7. RAGAS-Style Evaluation

RAGAS is commonly used to evaluate RAG pipelines with reference-based and/or reference-free metrics. For interviews, remember the distinction: retrieval metrics evaluate the retriever; faithfulness and answer relevancy evaluate the generated answer; context metrics evaluate the retrieved evidence.

Illustrative Python workflow

```
# Illustrative evaluation structure
sample = {
    "question": "What is the refund period?",
    "contexts": [
        "Refunds are available within 30 days of purchase.",
        "Refund requests require the original order number."
    ],
    "answer": "The refund period is 30 days."
}

# In a real project, pass a dataset like this to your
# chosen RAG evaluation framework and configure the
# appropriate evaluator/LLM/embeddings.
```

8. Interview Question: How do you evaluate a RAG system?

Strong answer: “I evaluate RAG at three levels. First, retrieval quality using Recall@K, Precision@K, MRR and NDCG. Second, end-to-end generation using faithfulness, answer relevancy and context-related metrics. Third, production health using latency, token cost, user feedback and failure rates. I maintain a representative evaluation dataset and run automated evaluations after changes to embeddings, chunking, retrievers, prompts or source data.”

9. Interview Question: Retrieval is returning old documents. What do you do?

Use metadata such as document version and effective date, apply metadata filtering where appropriate, add recency-aware ranking, and use a reranker or hybrid retrieval strategy. Then measure whether Recall@K, Precision@K and downstream faithfulness improve.

10. Retrieval Drift in Production

Retrieval drift means the retriever's behavior or quality changes as documents, embeddings, queries, chunking, or ranking logic change. Useful signals include declining Recall@K/Precision@K on a fixed evaluation set, lower context relevance, increased thumbs-down feedback, more “I don't know”/hallucination cases, and changes in query or document distributions.

Recommended monitoring loop:

- Keep a fixed golden evaluation set with questions and expected evidence.
- Run retrieval evaluation periodically and after major pipeline changes.
- Track top-K relevance and rank-quality metrics.
- Track faithfulness and answer relevancy for end-to-end behavior.
- Monitor user feedback, latency, token usage and cost.
- When quality drops, investigate chunking, embeddings, metadata filters, hybrid retrieval, reranking and source freshness.

11. Quick Interview Cheat Sheet

If interviewer asks...	Say...
Did we retrieve the right chunks?	Recall@K / Precision@K
Was the first useful result ranked highly?	MRR

If interviewer asks...	Say...
Is the ranking order good with graded relevance?	NDCG@K
Did the answer use only the evidence?	Faithfulness / groundedness
Did the answer address the question?	Answer relevancy
Was the retrieved context useful?	Context precision
Did retrieval capture enough evidence?	Context recall
Is production quality changing?	Golden-set evals + feedback + drift monitoring

Interview tip: Avoid saying “RAGAS checks whether retrieval is correct” as your only answer. Explain which metric answers which question, and distinguish retriever evaluation from generation evaluation.